

SQL QUERYING & SALES DATA INSIGHTS REPORT

Program: AnalystLab Africa Data Analytics Internship | Week 3 | Database: AnalystLabSalesDB

This secondary technical deliverable provides an advanced overview and analytical breakdown of the production-grade SQL scripts executed against the Kaggle global e-commerce B2B Sales transactions table using Microsoft SQL Server (T-SQL). The analysis covers metadata auditing, product line metrics, and core enterprise client valuation strategies.

1. Structural Metadata & Column Type Auditing

Query Operations:

- `CREATE DATABASE AnalystLabSalesDB;`
- `EXEC sp_columns Sales_Transactions;`

Why it Matters: Before running intensive numeric aggregations on an imported flat file, a structural metadata audit ensures column data types match mathematical expectations and highlights constraints (e.g., fields stored as text requiring active data type casting).

Data Observations & Insights: The table maps out key financial data points like `SALES` and `PRICEEACH` as float values, with operational transaction links mapped onto `ORDERNUMBER`. Crucially, the `ORDERDATE` is identified as a variable character string (`varchar`), signaling that any time-series analysis must actively apply `CAST(ORDERDATE AS DATETIME)` to avoid processing errors.

2. Product Line Velocity & Revenue Performance

Query Operations: Summarized gross revenue metrics grouped by category constraints using `GROUP BY PRODUCTLINE`, sorting the grid by cumulative sales volume.

Why it Matters: Category grouping exposes the structural revenue engines of the enterprise, allowing operations and inventory management teams to clearly prioritize product development budgets.

Data Observations & Insights: Your SQL output screen exposes **Classic Cars as the absolute blockbuster engine of the enterprise**, generating an outsized gross revenue total of **\$3,919,615.66** across 967 total orders and 33,992 units sold. This completely dwarfs the second-place category, **Vintage Cars**, which brought in **\$1,903,150.84**. Classic auto collectibles generate more than double the volume of vintage categories, representing the main business priority.

3. Customer Lifetime Value (LTV) & Account Concentration

Query Operations: Analyzed transaction velocity and cumulative investments per client by running:

- `SELECT TOP 10 CUSTOMERNAME, SUM(SALES) AS TotalSpent ... GROUP BY CUSTOMERNAME ORDER BY TotalSpent DESC;`
- `SELECT CUSTOMERNAME, COUNT(ORDERNUMBER) AS NumberOfOrders, SUM(SALES) ... ORDER BY NumberOfOrders DESC;`

Why it Matters: Account concentration tracking evaluates the firm's reliance on major B2B contracts. It separates high-frequency whale buyers from irregular or lower-value customer accounts.

Data Observations & Insights: Your query grid identifies **Euro Shopping Channel as the undisputed top enterprise customer account**, locking in a massive lifetime spend of **\$912,294.11** across a high-volume **259 separate orders**. The second most valuable client is identified as **Mini Gifts Distributors Ltd.**, driving a lifetime value of **\$654,858.06** over **180 total orders**. Strategic retention plans should prioritize these top two accounts, as they represent the core revenue baseline for the business.

4. Advanced Logical Implementations

Query Operations: Applied structured chronological and analytical expressions, establishing rolling analytical metrics over the dataset.

Why it Matters: Incorporating cumulative and indexed window queries (`RANK()` , `ROW_NUMBER()`) transitions basic static reporting into a dynamic analytical environment capable of tracking real-time growth indicators.

Data Observations & Insights: By leveraging window-framed functions like `SUM(SALES) OVER (ORDER BY ORDERDATE, ORDERNUMBER)` , the system calculates a seamless, unbroken row-by-row financial running total. This avoids data grouping blocks and gives management an exact look at corporate financial trajectory metrics at any given second.